Not Finished, But Abandoned

by

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> *It is customary for authors of academic books to include in their prefaces statements such as this: "I am indebted to ... for their invaluable help; however, any errors which remain are my sole responsibility." Occasionally an author will go further. Rather than say that if there are any mistakes then he is responsible for them, he will say that there will inevitably be some mistakes and he is responsible for them....*
>
> *Although the shouldering of all responsibility is usually a social ritual, the admission that errors exist is not — it is often a sincere avowal of belief. But this appears to present a living and everyday example of a situation which philosophers have commonly dismissed as absurd; that it is sometimes rational to hold logically incompatible beliefs.*
>
> — David C. Makinson (1965)

Above is the famous "preface paradox," which illustrates how to use the `wbepi` environment for epigraphs at the beginning of chapters. You probably also want to thank the Academy.

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ABSTRACT

FIXME: basically a placeholder; do not believe

I did some research, read a bunch of papers, published a couple myself, (pick one):

1. ran some experiments and made some graphs,

2. proved some theorems

and now I have a job. I've assembled this document in the last couple of months so you will let me leave. Thanks!

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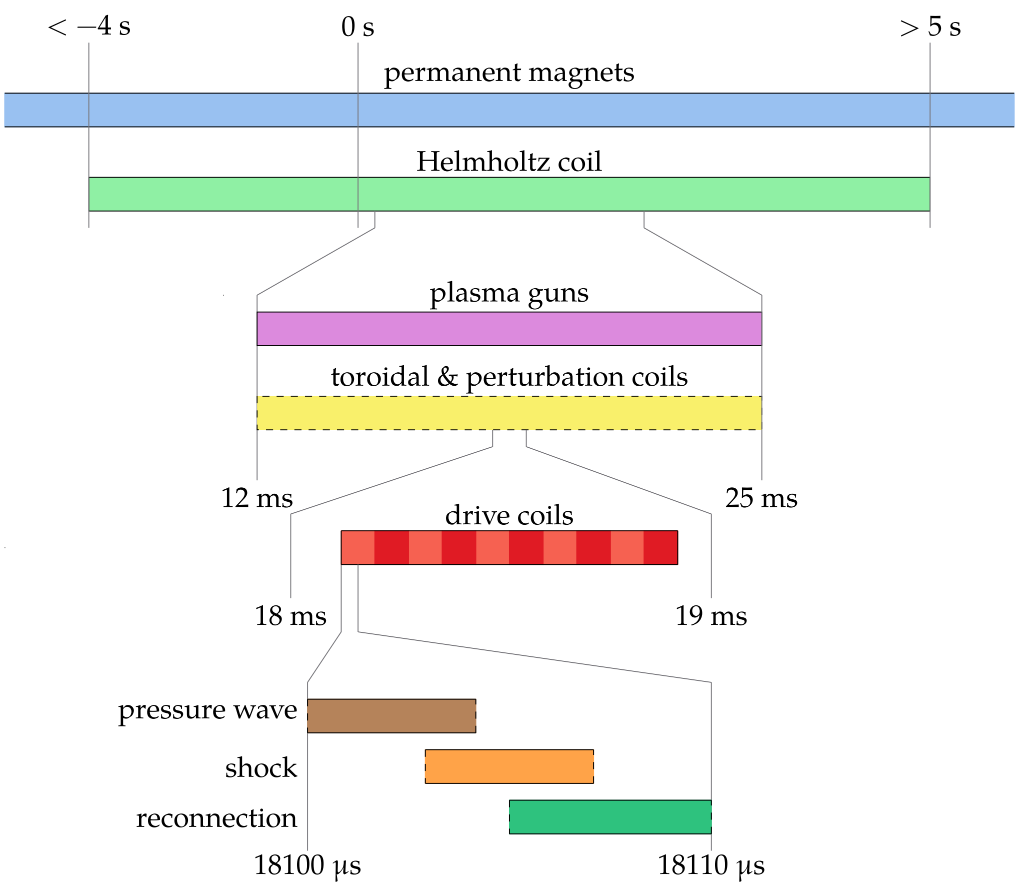

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3 MULTI-TIP LANGMUIR PROBE

3.1 What's a Langmuir probe

Hershkowitz: Hershkowitz (1989) Hutchinson: Hutchinson (2002) Olson: Olson (2020)

Plasma diagnostics make a broad phylogenetic tree ranging from the variety of plasma parameters they can measure to their size and complexity. In the Big Red Ball we prefer simplicity and cheapness to some of the wilder diagnostic varieties so a common probe that we use is a Langmuir probe.

3.1.1 Basic Construction

These Langmuir probes are quite basic in construction. At it's simplest, a conductor is placed in contact with the plasma and a voltage is applied relative to ground which we will call the probe voltage, V_{probe}. This will locally perturb the plasma and collect a current, I_{probe}, which can be seen in figure 3.1. If $V_{probe} < V_{plasma}$, an electric field is formed that repells electrons and attracts ions and the inverse for $V_{probe} > V_{plasma}$.

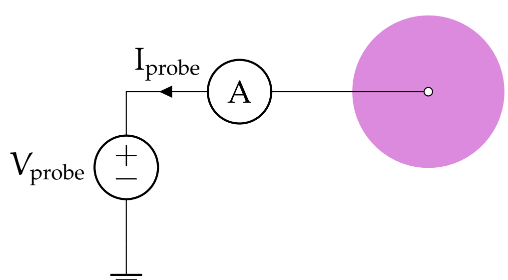

Figure 3.1: A circuit for a very simple Langmuir probe. The probe is biased to the probe voltage, V_{probe}, relative to ground by the power supply. The conductor (white circle) collects ions and electrons from the plasma (pink circle) causing a current to the probe, I_{probe}.

Langmuir probes have been developed to work in a very wide range of plasma parameter configurations that well encompass the plasma parameters TREX achieves (see table 1.2).

3.1.2 Extracting Plasma Parameters

The most challenging part of these probes is the conversion of measured current to plasma parameters. Since we are trying to get a local measurement of the plasma parameters and we are also perturbing the plasma locally by touching it with a probe, we need to create a theory that correctly takes into account that perturbation such that we can get the plasma parameters. This often consists of a fitting procedure where the difference between the theoretical and experimental I-V curves is minimized.

Additionally, many a probe theory is purpose built for plasmas with no magnetic field. As we are studying magnetic reconnection in which there are in fact magnetic fields, this could change the I-V characteristic in unexpected ways. We'll discuss below why these effects are not very impactful and in future sections describe the theories we use for extracting the plasma parameters.

3.1.3 Plasma Perturbation

A problem often found in highly magnetized plasma devices is the long range perturbation to the plasma by the probe. In the Big Red Ball there are times where we have a relatively strong field and times when we don't so let's examine both cases.

3.1.3.1 Unmagnetized Perturbation

Let's examine this from an unmagnetized starting point. In this configuration, the probe collects nearly isotropically to an extent limited by the Debye length which creates a sheath around the probe and operates to shield the electric field caused by the difference between V_{probe} and V_{plasma}. This length scale is very short and plasma can easily refill the sheath in this unmagnetized system.

3.1.3.2 Magnetic Field Effects

Now in a highly magnetized system (gyroradius far smaller than probe length scale) the plasma interacting with the probe comes from a flux tube approximately the size of the probe area projected into the magnetic field direction. Because of this, the probe can start "emptying out" the flux tube because of the difficulty of plasma to diffuse across field lines into the flux tube. This means the extend of the probe perturbation is far longer.

3.1.3.3 Perturbations in our System

We are quite lucky in that our system geometry reduces the impacts of flux tube emptying and the plasmas we are studying are often relatively unmagnetized.

3.1.3.3.1 Geometric Considerations The TREX experiment is (often) cylindrically symmetric and our probes are placed in different toroidal planes (see figure 2.6 for coordinate system reference). This means that without any toroidal field, any emptying only occurs in the plane of the probe which may still affect the probe I-V characteristic but is not affecting other probes measurements in a significant manner. When we do have a toroidal field, we also often have non-uniform B_r and B_z which means the flux tube passes through many different plasma parameters. This may help to average out any flux tube emptying because at different points along the flux tube more plasma may be able to enter the tube than just what is expected at the probe causing a reduced perturbation length.

3.1.3.3.2 Magnetic Field Considerations As we have magnetic fields in our system, we do need to worry about these flux tube effects. Fortunately, our magnetic field are relatively small and in the region we are often most interested in (where reconnection is occurring) the magnetic field is far smaller than the initial background field. We characterize this smallness by the ratio of the gyroradius to the radius of the probe area which is on the order of ??? (see 3.1).

3.2 Hardware

There has been a long and storied history of probe building on the Big Red Ball expecially for the development of multi-tip Langmuir probes. For some more information about our probes, see Olson (2020).

3.2.1 Physical Constraints

Our Langmuir probes need to satisfy a number of requirements.

- **Length**: The probe is attached to a port on the surface of the machine. This means it starts at a spherical radius of 1.5 m and must be inserted from there. Thus the insertable length should be fairly long and must be stable. This is described in 3.2.2.

- **Material**: We want our probe to survive the plasma exposure and not affect our vacuum pressure via offgassing. We also don't want to have to worry about secondary electron emission. We discuss this in 3.2.3.

- **Probe Size**: We want to measure the plasma locally so our probe body needs to be small relative to the plasma gradients. It also needs to be able to fit through ball joints that can be mounted to the machine

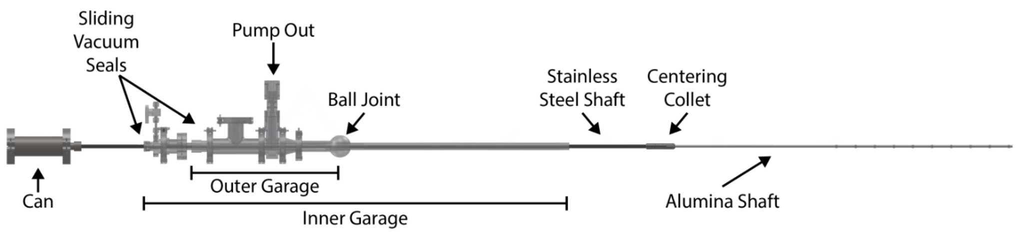

Figure 3.2: Three stage probe shaft with an outer and inner garage and the main stainless steel shaft that connects to the probe.

that have a radius of ???. These considerations are taken into account in 3.2.3.

3.2.2 Shaft Components

For our probe to reach everywhere we want and also actually be in the position we desire we need a long shaft with a sufficient amount of support to be accurate to a few centimeters. Why a few centimeters? Our plasma dynamics are reproducible but not to such an extreme that a centimeter is noticeably different. Also it's difficult with the probe length to achieve too much better.

3.2.2.1 Garages

As can be seen in figure 3.2, the probe shaft is a multistage system. Each stage consists of some tubing where connections between different stages are made using quick-connect high-vacuum fittings (like part 3486N32 from McMaster-Carr).

We mount the probe to the machine using a flange on the outer garage or a ball joint (as seen in the figure). Then we can move each garage independently to insert the probe a desired length.

3.2.2.2 Shaft materials

We need to sufficiently support our probe and the shafts themselves so we use stainless steel until we enter the plasma region in which we change to alumina.

For the shafts, stainless steel is vacuum safe, tolerant to plasma interaction, and can support the weight of the structure easily without excessive flexing. Unfortunately, as a conductor, it will act as a long Langmuir probe that will create a ray in the machine at a single potential. This will perturb the plasma much more significantly than using an insulator.

To that end we use alumina, a high temperature ceramic who's surface will accumulate charge such that it'll be left at the floating potential (the potential at which the ion and electron currents balance). This helps to perturb the plasma less as there is no net current to the shaft and thus the plasma should (hopefully) be less perturbed.

Right near the end of the probe shaft, we use stainless steel tubing to connect from the alumina to the probe body. This is to reduce the overall size of the shaft in perturbing the plasma at the probe and is only for 10 cm of length.

3.2.2.3 Centering Collet

Each shaft/garage, is held in place by the sliding vacuum seal that connects it to it's larger garage. Unfortunately this gives only a single point of contact for making the shafts concentric. This would cause excessive drooping of the probe and may be so bad as to cause the probe to hit the edge of the machine when being inserted/removed.

To fix this issue, we have used centering collets that allow the inner shaft/garage to move while being held in place by the outer garage. These can be both machined or 3D printed. When 3D printed we can make these to also hold the o-rings so they are double-acting.

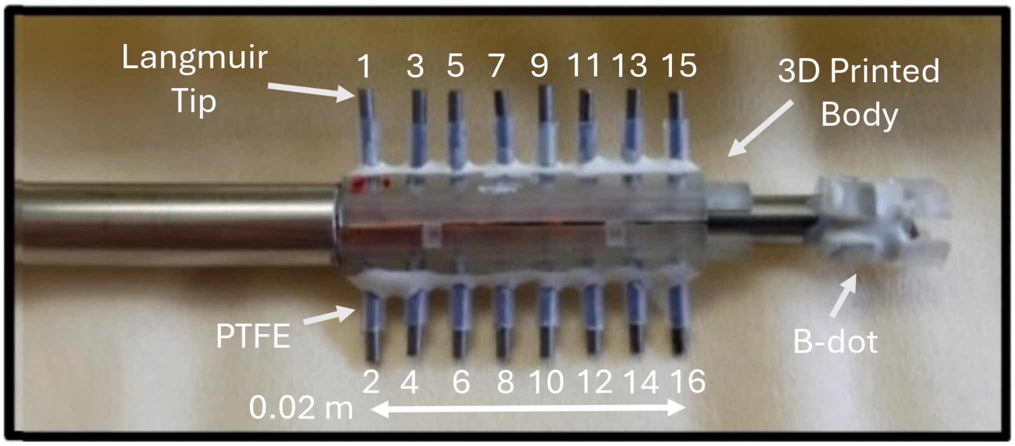

Figure 3.3: Picture of the end of the multi-tip Langmuir probe. Image from Gradney (2025).

3.2.2.4 Feed-thru

We need some way to extract the signals from the probe tip. Because of the construction of our probe and for simplicity purposes, the probe shaft is at vacuum pressure. Thus we attach a feed-thru to the end of the can from figure 3.2. This is constructed using two DB-25 hermetically sealed connectors where we solder the wires directly to the vacuum side of the can and then can easily move and connect the probe externally.

3.2.3 Tip Components

3.2.3.1 Probe body

Our probe does not need to withstand extreme heat fluxes and our vacuum conditions don't need to be perfect so we are lucky enough to be able to use a 3D printed probe body.

This is constructed using stereolithography which gives a sufficient resolution for the probe. Additionally the plastic used holds up well in vacuum. The body must connect to two different diameter stainless

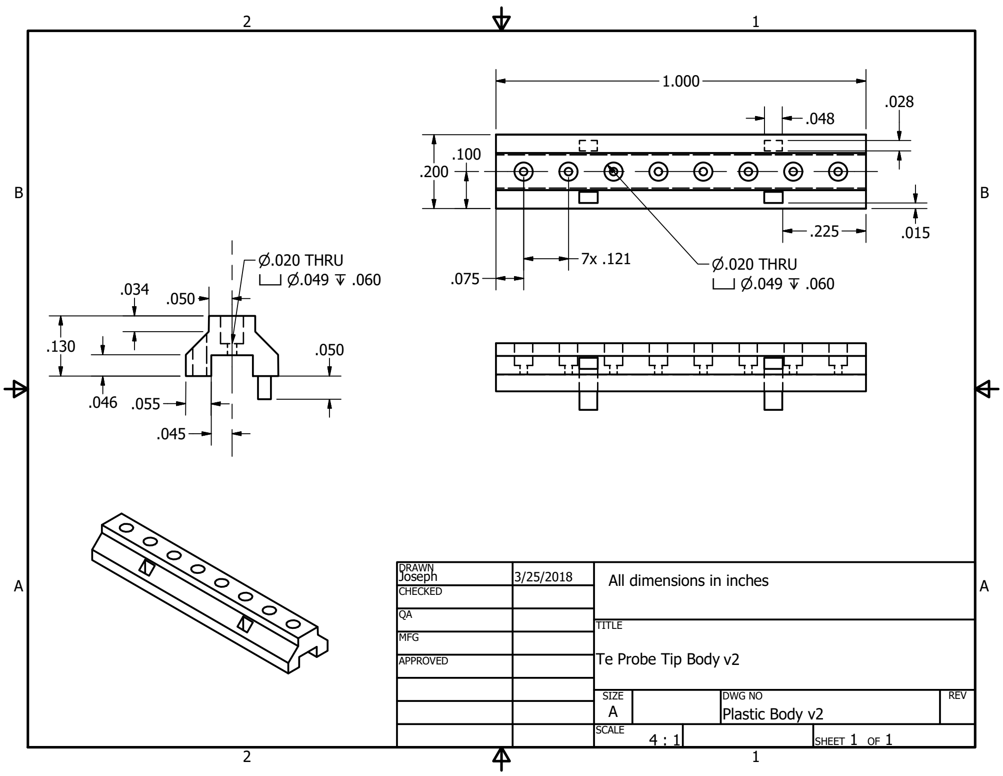

Figure 3.4: Probe body from engineering sketch. Created by Joe Olson.

steel tubes (as can be seen in figure 3.3). Thus we use two annuli whose inner/outer radii match the tubing.

The tubing exists to connect the probe to the main shaft holding it up and also a small B-dot coil which is discussed in 3.2.4.

3.2.3.2 Tip material + size

To collect current, we use 16 separate tips which each sample a point on the IV curve. We make these tips out of Molybdenum which requires us to connect the tips to the signal wire using a friction fit.

3.2.3.2.1 Why use Mo Even though Molybdenum has it's difficulties with connections, it has a high work function which helps to reduce secondary electron emission which can change the shape of the IV-characteristic.

3.2.3.2.2 Size considerations We expect the current to the tips to be on the order of 1 kA/m^2. We measure the current across a 'sense' resistor (as discussed thoroughly in 3.3.2) but we don't want to have to use high power resistors for this. Thus we choose our current to be on the order of 10 mA which is feasible by having a tip area on the order of 10 mm^2. This amount of current also gives a reasonable signal-to-noise ratio.

Unfortunately, the probe is not all that cylindrical which means that the idealized probe theories we use are not as precise as we may have hoped. Future probe designs could use longer and smaller diameter tips to avoid this issue.

3.2.3.3 Tip Shielding

We protect each tip from the base of the probe body using PTFE/teflon. Teflon isn't great in a vacuum environment because of offgassing but does the trick and is cheap and easy to use. After inserting the tip into the probe body we push the teflon on and apply a small amount of vacuum-safe epoxy at the base to hold the shielding in place.

3.2.3.3.1 Why shield at all? We could potentially shorten each tip so the collecting area extends all the way to the probe body. Unfortunately because of the increased size of the body this can cause shadowing effects where tips on one side of the probe have a different IV-characteristic than those on the other. This 'shadowing' is caused by plasma flow parallel to the tip direction whcih gets intercepted by the probe body.

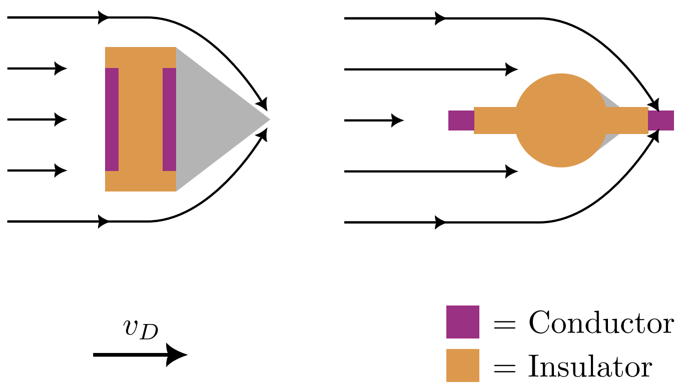

Figure 3.5: Simplified cartoon of the expected mach effect for the multitip probe. Left panel inspired by Merlino and D'Angelo (1987), Fig. 10.

3.2.3.3.2 First order mach effects to avoid Another reason to shield the probe is to avoid mach effects that would cause different IV-characteristics for tips on opposite sides. As can be seen in figure 3.5, in an ideal mach probe the ions are prevented from easily accessing the "shadowed" side of the probe (right side) because of their inertia. If we shrink the probe body and increase the distance the tip is from the probe the mach effect is reduced because ion inertia is not sufficient to prevent collection. This is only approximate though and we sometimes do see effects that could be caused by shadowing. For an in depth discussion of mach probes we refer the interested reader to Chung (2012).

3.2.4 B-dot Coils

Since we don't know the exact location of the probe and the plasma is not perfectly reproducable, it's especially useful to have measurements we can compare to other diagnostics to locate the probe in space and around the reconnection region. To do this we use hand wound three axis Bdot coils that can measure the change in magnetic field near the tips.

3.2.4.1 Size Considerations

Through past trial and error and many a day spent hand winding coils for past students, we've found a reasonable vector area magnitude to use for the B-dot coils. During reconnection the largest $\partial_t \mathbf{B}$ is on the order of 10 kT/s. Thus, from Faraday's law,

$$\mathcal{E} = \mathbf{A} \cdot \partial_t \mathbf{B} \tag{3.1}$$

where $\mathcal{E}$ is the electromotive force and $\mathbf{A}$ is the vector area of the coil Jackson (1999). This means that for a signal of 1 V we are looking for a cross-sectional area of 10^{-4} m^2.

Using a 3D-printed B-dot body (seen in right in figure 3.3) we wrap wire around a square shaped center six times giving an area of $\sim 5{\times}10^{-5}$ m^2. Although smaller than that required for a 1 V measurement, this makes sure the signal doesn't become too large should some kind of resonance exist in the circuit.

3.2.4.2 Noise Reduction

Since common mode noise can relatively easily couple to the coils, we use twisted pair for each wrapping direction. This gives us two circuits that have nearly opposite vector area, $\mathbf{A}_1 \approx -\mathbf{A}_2$, such that we can subtract the two measurements and thus remove common mode contributions. We will discuss this more in 3.3.3.4.

3.2.5 Length Scales

Table 3.1 contains all information that helps us decide on our Langmuir probe theory and signal interpretation. There are a few key features to note that let us validate the use of finite area B-dot coils and multiple tips.

Name	Symbol	Equation	Value
Tip radius	r	—	5e-4
Tip length	l	—	2e-3
Neighboring tip separation	—	—	2e-3
Across probe tip separation	—	—	1.5e-2
B-dot coil side length	—	—	3e-3
Debye length	λ_D	$\sqrt{\frac{\epsilon_0 T_e}{ne}}$	3e-5
Ion Gyroradius	r_i	$\frac{m_i\sqrt{2m_i\bar{v}_i}}{eB}$	6e-1
Electron Gyroradius	r_e	$\frac{m_e\sqrt{2m_e\bar{v}_e}}{eB}$	1.5e-1
Ion mean free path	λ_i	$\frac{\bar{v}_i}{\nu_{i,i}}$	9e-4
Electron mean free path	λ_e	$\frac{\bar{v}_e}{\nu_{e,e}}$	4.5
Electron energy relaxation length	λ_ε	* Eq. 5	4.5
Sheath thickness	h	$\sim \lambda_D$	3e-5

Table 3.1: Length scales of the multitip probe. All lengths are in meters and are approximate. For the interested reader and for the energy relaxation length, please refer to Demidov et al. (2002) for Langmuir probe relevant physics of these scales. These parameters were calculated with $n = 10^{18}$ m^{-3}, $T_e = 20$ eV, $T_i = 0.1 \times T_e$, and $B = 10^{-4}$ T. For the mean free paths we have chosen the frequency that gives the shortest length.

Our ion and electron gyroradii are both relatively large compared to the probe tips and B-dot coils. This means that if we assume the plasma is relatively uniform in a flux tube, the entire probe is sampling a quite uniform plasma.

3.3 Circuitry

The multitip Langmuir probe consists of a number of circuits that work together to give us some nice measurements. Here we enumerate their operation methods, upgrades, signal extraction, and calibration techniques.

3.3.1 Bias Source

We have thus far had two different circuits that we've used for biasing each tip relative to the laboratory ground. The first, made by Jan Egedal while at VTF is manually controlled, while the second, made by me is remotely operated. We'll discuss how they work and their pros and cons.

3.3.1.1 Manually Controlled

Our previous setup consisted of multiple electrolytic capacitors who's voltage is set by a series of resistors that split the voltage between the negative power supply (V_-), ground, and the postivie power supply (V_+).

3.3.1.1.1 Principles of operation Figure 3.6 shows a 4 tip biasing circuit (our circuit uses 16 tips but we have simplified it). The 'rails' (which are the maximum and minimum voltage of the circuit) are set by power sources at V_+ and V_- which are biased relative to the ground input connection labeled `Jgnd`. A series of resistors splits the voltage between the rails and we choose one of the tips to be at laboratory ground. This lets us set the maximum and minimum voltage along with setting how many tips are positively/negatively biased. Also, moving the ground changes the voltage separation between tips so we can get higher resolution. The voltage separation charges the respective capacitors for each tip to a bias voltage, $V_b = \texttt{Vb}$. The capacitors exist to keep the bias voltage of each tip nearly constant during an experimental shot.

Now let's calculate the bias for each tip. Let $i, j \in \{1, \ldots, N\}$ be the index of the tip and the index of the ground connection respectively. Then,

$$V_{bi} = \begin{cases} \frac{V_+}{j}(j-i) & \text{if } j \geqslant i \\ \frac{V_-}{j}(j-i) & \text{if } j < i \end{cases} \tag{3.2}$$

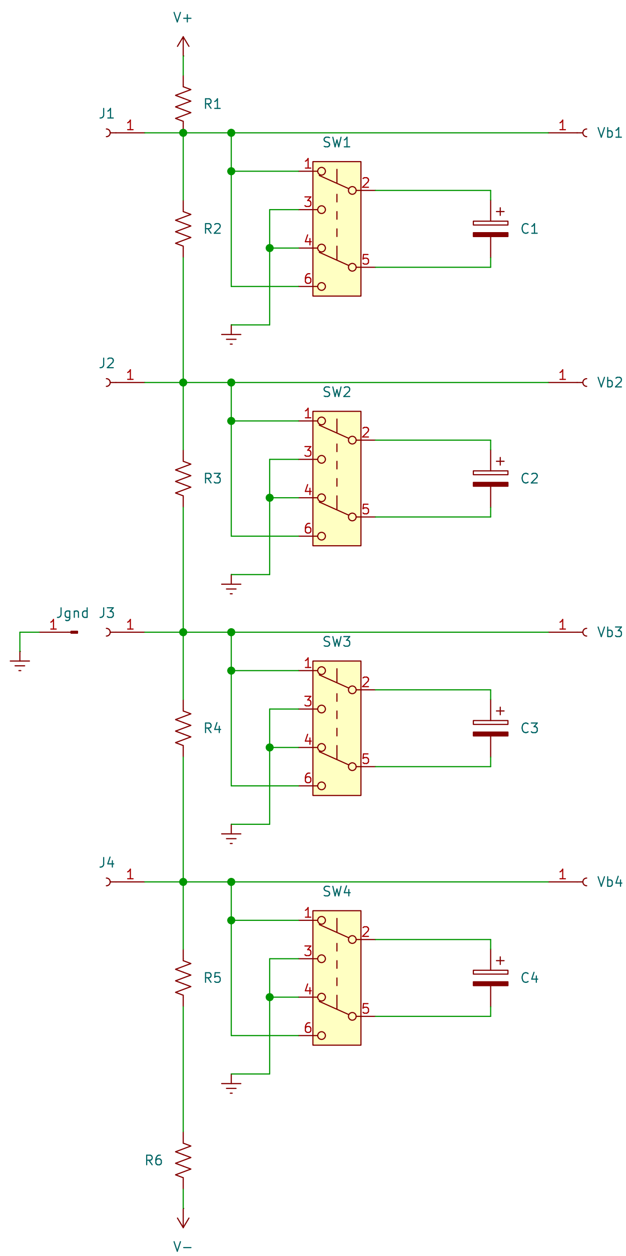

Figure 3.6: Old biasing circuit for the multitip Langmuir probe.

Figure 3.7: Image of old setup

Figure 3.8: Circuit diagram of new setup.

Figure 3.9: Image of new setup

Table 3.2: Bias circuit component values

3.3.1.1.2 Problems - Had to be manually controlled for drastic bias changes - Wires often contacted each other or parts would lose contact

3.3.1.2 Remote Controlled

3.3.1.2.1 Principles of operation - Equation to use for calculation

3.3.1.2.2 Problems - Destroyed a power supply - A resistor was designed and thus burnt out

3.3.1.2.3 Calibration

3.3.2 Probe Circuit

3.3.2.1 Principles of operation

3.3.2.1.1 Plasma draws current

3.3.2.1.2 Voltage drop across sense resistor

3.3.2.1.3 Measure high frequency voltage drop

Figure 3.10: Simple circuit diagram

Figure 3.11: Image of old circuit board

Figure 3.12: Image of new circuit board

Table 3.3: Probe circuit component values

3.3.2.2 Single-ended measurement

3.3.2.3 Differential measurement

3.3.2.4 Circuit Solution

3.3.2.4.1 How do we solve this

3.3.2.4.1.1 Kirkoff's laws

3.3.2.4.1.2 System of equations

3.3.2.4.1.3 SymPy

3.3.2.4.2 equations

3.3.2.4.3 Assumptions

3.3.2.5 Calibration

3.3.2.5.1 Calibrating in isat

3.3.2.5.2 Calibrating on the exponential

3.3.2.5.3 Bayesian calibration

3.3.2.5.4 Calibration using pressure wave

Table 3.4: Chen parameters

3.3.3 Bdot coils

3.3.3.1 In-shaft circuit

3.3.3.1.1 Why do we have that?

3.3.3.1.2 Where is it placed approximately?

3.3.3.2 Circuit model

3.3.3.3 Calibration

3.3.3.3.1 Directional

3.3.3.3.2 Transfer Function

3.3.3.4 Signal Interpretation

3.4 Plasma Theory

3.4.1 Simple Plasma Theory (Hutchinson)

3.4.1.1 Derivation

3.4.1.2 Pros & Cons

3.4.2 BRL

3.4.2.1 Idea behind it

3.4.2.2 Approximations of it

3.4.2.2.1 Planar probe

Figure 3.13: What does an electric field do to the tip measurements diagram.

Figure 3.14: Example plot where E splitting tips

3.4.2.2.2 Chen Approximation

3.4.2.2.3 Even more approximation

3.4.3 Effect of electric fields

3.4.3.1 Where does the electric field come from

3.4.3.1.1 Majority inductive

3.4.3.1.2 Some static

3.4.3.1.2.1 What is the expected static magnitude

3.4.3.2 What does it do to the tip measurements

3.4.3.2.1 Theoretical understanding

3.4.3.2.2 Physical reality

Figure 3.15: Diagram of local minimums

3.4.4 Minimum measurable density

3.4.4.1 What's the debye Length

3.4.4.2 Sheath scale

3.4.4.3 Tip-tip sheath overlap

3.5 Curve Fitting

3.5.1 Least squares Fitting

3.5.1.1 Benefits (easy)

3.5.1.2 Issues (not representative of error)

3.5.2 ODR

3.5.2.1 What does error look like in our signals

3.5.2.2 How does ODR take that into account

3.5.2.3 How to apply parameter bounds

3.5.2.3.1 Our method

3.5.2.3.2 Other options

Figure 3.16: Example fit using many shots, multiple local parameter fits, and errors

Figure 3.17: Plot of isat currents from BRL

3.5.3 Monte-carlo Fitting

3.5.3.1 What kind of fits do we get if we just trust a local minimum

3.5.3.2 How do we sample initial conditions

3.5.3.3 More robust methods for getting global minimums

3.5.4 Combining multiple shots

3.5.4.1 Using some time points from each shot

3.5.4.2 Combining all IV points and fitting that

3.5.4.3 Problems

3.5.4.3.1 Timing alignment We'll go into this more in the PA section.

3.5.4.3.2 Variability of plasma parameters

3.6 Results

What is the benefit of doing all this work?

Figure 3.18: Radial chord of initial density with Hutchinson vs BRL.

Figure 3.19: Comparison of expected E from Ohm's law and measured E.

3.6.1 Measurement of low densities

3.6.1.1 How does Hutchinson vs BRL treat ion saturation

3.6.1.2 What are the experimental predictions that Hutchinson vs BRL have

3.6.1.3 How does that change the results

3.6.2 Noise resilience

Not sure how best I should show this.

3.6.3 Fits more accurate

- Each fit uses many data points
    - Don't have to arbitrarily cutoff data points

3.6.4 Extracting electric fields

3.6.4.1 Where do we expect electric fields to be

3.6.4.2 Do we see them there

4.1 How can we measure Pressure Anisotropy

4.1.1 Thomson Scattering

4.1.1.1 Pros

4.1.1.2 Cons

- Requires a lot of playing around with optics
- Size of machine
- Don't have much in-lab expertise
- Getting all the photons is tough
- Don't have optics stuff setup

4.1.2 Directional Langmuir Probe

4.1.2.1 Pros

- Simplicity
- Cost
- Mobility

4.1.2.2 Cons

- Difficult to interpret results precisely (luckily only need relative stuff)
- Can't perfectly model
- Requires a lot of playing around with circuits

Figure 4.1: Picture of feed thru along with drawings of cross-sections.

Figure 4.2: Picture of end of probe

Figure 4.3: CAD diagram of full body

4.2 Hardware

4.2.1 Shaft Components

4.2.1.1 Garages

4.2.1.2 Shaft materials

4.2.1.3 Centering Collet

4.2.1.4 Feed-thru

4.2.2 Tip Components

4.2.2.1 Probe body

4.2.2.1.1 3D printed

4.2.2.1.2 Both sides fit together

4.2.2.2 Tip material + size

4.2.2.2.1 Shell material

4.2.2.2.2 Core material

Figure 4.4: CAD diagram of probe tip

Table 4.1: Length scales of probe

Figure 4.5: Simple circuit diagram of how bias works.

4.2.2.3 Tip Shielding

4.2.2.3.1 Shell Shielding

4.2.2.3.2 Core Shielding

4.2.3 Bdot Coils

Same as Te probe

4.2.4 Connecting Bdot - Probe - Shaft

4.2.5 Length Scales

4.3 Circuitry

4.3.1 Bias Source

4.3.1.1 Principles of operation

4.3.1.1.1 Charging

4.3.1.1.2 Discharging

4.3.1.1.3 Swapping polarity

4.3.1.2 How do we do this

4.3.1.2.1 Cannibalization

Figure 4.6: Full circuit diagram of how bias works.

Figure 4.7: Image of power supply setup

4.3.1.2.2 Discharging

4.3.1.2.3 Safety Diodes for discharging the wrong way.

4.3.1.2.4 Cap arcing

4.3.1.2.4.1 What happened

4.3.1.2.4.2 Solution

4.3.1.2.5 Reducing voltage drop due to inductance

4.3.1.2.5.1 Twisting Wires

4.3.1.2.5.2 Adding cap next to board

4.3.1.2.6 Reversing polarity

4.3.1.2.6.1 Why we break relay

4.3.1.2.6.2 How do we fix this problem

4.3.1.2.7 LabView (ew)

4.3.1.2.7.1 Controls CRIO

4.3.1.2.7.2 Countdown if too high of voltage for swapping

Table 4.2: Bias circuit component values

Table 4.3: Probe circuit component values

Figure 4.8: Schematic of grounding diagram

4.3.1.3 Calibration

4.3.1.3.1 Calibrating power supply output

4.3.1.3.2 Calibrating cap bank output

4.3.2 Probe Circuit

4.3.2.1 Single-ended measurement

4.3.2.1.1 TREX cells

4.3.2.1.2 Why was this bad

4.3.2.1.3 How long did we spend on this :(

4.3.2.2 Differential measurement

4.3.2.2.1 Parallel circuitry

4.3.2.3 Figuring out grounds

4.3.2.3.1 Types of grounds on the ACQ

4.3.2.3.2 Routing grounds

Figure 4.9: Image of swapping box

Figure 4.10: Diagram of swapping

Figure 4.11: Time frame for core movie of IV with separation between relay

4.3.2.4 Swapping current paths

4.3.2.4.1 Why do we want to do this

4.3.2.4.1.1 Takes care of small differences in resistors

4.3.2.4.2 Circuitry to do this

4.3.2.4.2.1 Only swap core stuff

4.3.2.4.2.2 Many relays

4.3.2.4.2.3 Remote control

4.3.2.4.3 How does this change results

4.3.2.5 Circuit Solution

4.3.2.5.1 Solution

4.3.2.5.2 Assumptions

4.3.2.6 Calibration

4.3.2.6.1 Calibrating in isat

Figure 4.12: Circuit diagram of probe circuitry

Figure 4.13: Image of probe circuitry

Figure 4.14: Probe signals showing coupling

4.3.2.6.2 Calibrating on the exponential

4.3.2.6.3 Rotation calibration

4.3.2.7 The path to ground

4.3.2.7.1 Why do we have to care

4.3.2.7.2 How do we set it

4.3.2.8 Capacitive coupling

4.3.2.8.1 Where does it come from

4.3.2.8.2 Where do we see it

4.3.2.8.3 Magnitude of effect

4.3.2.8.4 Removing effect

Figure 4.15: Figure of theoretial plasma distributions and JV curves

4.4 Plasma Theory

4.4.1 A full model

4.4.1.1 Plasma Distribution

4.4.1.2 Ion particle flux

4.4.1.2.1 Mach effects

4.4.1.3 Electron particle flux

4.4.1.3.1 Minimum velocity

4.4.1.3.2 Area

4.4.1.3.3 Particle Flux

4.4.1.4 Theoretical JV curves

4.4.1.4.1 Reading radar plots

4.4.1.4.1.1 Radar axis

4.4.1.4.1.2 Solutions

4.4.2 Individual tip fitting

4.4.2.1 Why do this instead

4.4.2.2 Methods

4.4.2.2.1 Aforementioned methods

Figure 4.16: Simple drawing of what might happen to ions

4.4.2.2.2 Exponential

4.4.2.2.2.1 Ions don't hit core

4.4.2.2.2.2 Take care of this by allowing free parameters

4.4.2.2.2.3 Converting from plasma parameters to exponential

4.4.2.2.2.4 Converting back

4.4.2.2.2.5 Taking care of errors

4.4.2.3 Extracting plasma parameters

4.4.2.3.1 Spherical harmonics

4.4.2.3.2 Error bars

4.4.2.3.2.1 Monte-carlo

4.4.2.3.2.2 Other options

4.4.2.3.2.3 Why can we ignore non-bi-maxwellian features

4.5 Curve Fitting

We will use ODR like mentioned in Te section.

Figure 4.17: Comparison of alignment using Bz = 0 or bdot cross correlation

4.5.1 Combining multiple shots

4.5.1.1 Timing alignment

4.5.1.1.1 Bz = 0 alignment

4.5.1.1.1.1 Aligns where we expect Anisotropy

4.5.1.1.1.2 Depends heavily on prior bdot data

4.5.1.1.2 Bdot cross correlation

4.5.1.1.2.1 Still requires smoothing to get rid of initial oscillation impact

4.5.1.1.3 Result comparison

4.5.1.2 Averaging of core data

4.5.1.2.1 Weighted average of relay on/off

4.5.2 Fitting shell

4.5.2.1 Use all shell data

4.5.2.2 Fit using BRL

4.5.2.3 Fits pretty good

4.5.3 Fitting core

4.5.3.1 Fitting using full routine

4.5.3.1.1 Slow

4.5.3.1.2 Model not perfectly representative

4.5.3.2 Fitting tips independently

4.5.3.2.1 Individual fitting method

4.5.3.2.2 Holding Vprobe constant

4.5.3.2.3 Exponential fitting method

4.5.3.2.4 Spherical harmonics

4.6 Results

Figure out what my results are once I have some reasonable results

4.6.1 Drive cylinder

4.6.1.1 What did we try

4.6.1.2 Why didn't we see pressure anisotropy

4.6.1.2.1 In the bdot data

4.6.1.2.2 In the PA data

4.6.2 3 coil

4.6.2.1 How much anisotropy should we expect

4.6.2.2 What setups did we try

4.6.2.3 Old results of just comparing radar plots

4.6.2.4 New results using independent fitting

5 CONCLUSION

5.1 Summary of thesis

5.1.1 Using pressure waves to measure density

5.1.2 Upgrading langmuir probe analysis and new possibilities

5.1.3 Using directional langmuir probes for measuring pressure anisotropy

5.2 Future work

5.2.1 Pressure wave

5.2.1.1 Aforementioned future work

5.2.1.2 Extracting cross-field diffusion

5.2.2 Langmuir Probe

5.2.2.1 Prepared to analyze lots of cool data

5.2.2.2 Extract out electric field

5.2.3 PA Probe

5.2.3.1 Try more configurations

5.2.3.2 Map out full pressure anisotropy during shot

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